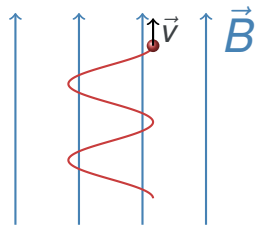
Quantum Motion in a Uniform Magnetic Field



Ali EL AZDI (IN BA5)

Why study this problem?

- **Theoretical Foundation**: One of the few exactly solvable interactions in Quantum Mechanics
- **Condensed Matter**: Describes electrons in metals and semiconductors (2D electron gas)
- **Quantum Hall Effect**: Explains macroscopic quantum phenomena and resistance standards

Background: What we need to know

A particle with charge q moving at velocity $\vec{v}$ in an electromagnetic field is subject to the general **Lorentz Force**:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

E the electric field, B the magnetic field, q the charge of the particle and $\vec{v}$ the velocity of the particle.

Example: Classical Motion

Setup Conditions:

- **System:** Particle $q < 0$ confined to xy-plane
- **Field:** Static uniform $\vec{B} = B\hat{z}$ ($E = \frac{dB}{dt} = 0$)
- **Motion:** Cyclotron orbit $r = \frac{mv}{|q|B}$

Example: Classical Motion

What are the quantum energy levels of a charged particle in a uniform magnetic field?

The Vector Potential

Since $\nabla \cdot \vec{B} = 0$, we can write: $\vec{B} = \nabla \times \vec{A}$

Gauge Freedom: $\vec{A}$ is not unique $\vec{A}' = \vec{A} + \nabla\chi$.

We choose the **Landau Gauge** for $\vec{B} = B\hat{z}$:

$$\vec{A} = (0, Bx, 0)$$

Deriving the Classical Hamiltonian

Energy and Momentum

- Total Energy: $E_{total} = \frac{1}{2}m\vec{v}^2 + q\phi$
- Canonical Momentum: $\vec{p} = m\vec{v} + q\vec{A}$

$$H = \frac{1}{2m}(\vec{p} - q\vec{A})^2 + q\phi$$

Verification: Hamilton's equations recover the Lorentz force:

$$\dot{\vec{r}} = \nabla_{\vec{p}} H, \quad \dot{\vec{p}} = -\nabla_{\vec{r}} H$$

Quantum Mechanics Setup

Canonical Quantization: promote variables to operators.

- Position: $\vec{r} \to \hat{\vec{r}}$
- Momentum: $\vec{p} \to \hat{\vec{p}} = -i\hbar\nabla$

Substitute into Classical Hamiltonian:

$$H = \frac{1}{2m}(\vec{p} - q\vec{A})^2 \quad \longrightarrow \quad \hat{H} = \frac{1}{2m}(\hat{\vec{p}} - q\vec{A}(\hat{\vec{r}}))^2$$

Hamiltonian under Landau Gauge

- Schrödinger Equation: $\hat{H}\,|\psi\rangle = E\,|\psi\rangle$

Using $\vec{A} = (0, Bx, 0)$:

$$\hat{H} = \frac{1}{2m}\left[\hat{p}_x^2 + (\hat{p}_y - qBx)^2 + \hat{p}_z^2\right]$$

The Schrödinger equation becomes:

$$\frac{1}{2m}\left[\hat{p}_x^2 + (\hat{p}_y - qBx)^2 + \hat{p}_z^2\right]\psi = E\psi$$

Solving the Equation

Since $\hat{H}$ is independent of y and z,
$p_y = \hbar k_y$, $p_z = \hbar k_z$ are conserved.
Ansatz:

$$\psi(\vec{r}) = e^{i(k_y y + k_z z)} \phi(x)$$

Definitions

$$\omega_c = \frac{|q|B}{m}$$

$$x_0 = \frac{\hbar k_y}{qB}$$

This reduces the problem to 1D:

$$\left[\frac{\hat{p}_x^2}{2m} + \frac{1}{2} m \omega_c^2 (x - x_0)^2 \right] \phi(x) = \left(E - \frac{\hbar^2 k_z^2}{2m} \right) \phi(x)$$

Result: Landau Levels

We recognize a **Harmonic Oscillator's** Hamiltonian!

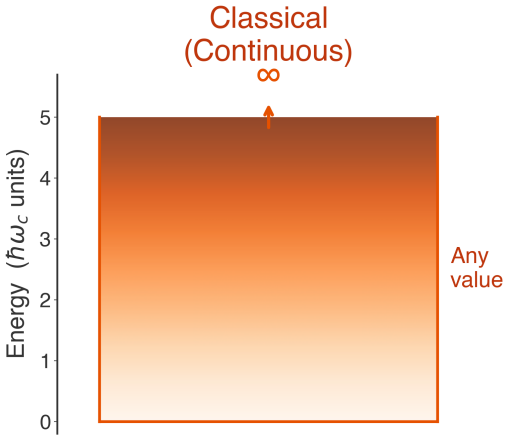
Energy eigenvalues:

$$E_{n,k_z} = \hbar\omega_c \left(n + \frac{1}{2} \right) + \frac{\hbar^2 k_z^2}{2m}$$

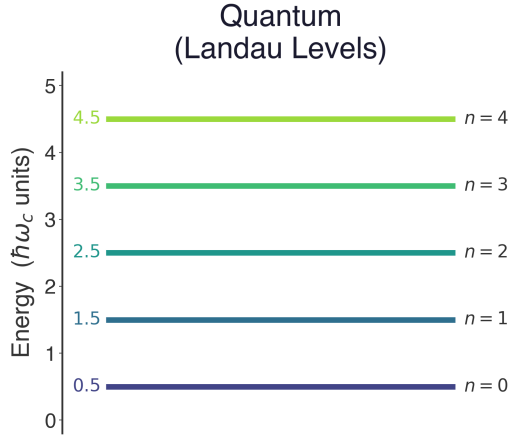
for $n = 0, 1, 2, \ldots$

- n is the harmonic oscillator's excitation level, quantized cyclotron orbits (**Landau Levels**)

Energy Levels



$E = \frac{1}{2}mv^2 + \frac{p_z^2}{2m}$ (any $E \geq 0$)



$E_{n,p_z} = \hbar\omega_c\left(n + \frac{1}{2}\right) + \frac{p_z^2}{2m}$

The Complete Wavefunction

- Center of oscillation $x_0 = \frac{\hbar k_y}{qB}$ depends on wavenumber k_y
- ϕ_n is the harmonic oscillator wavefunction

Full solution:

$$\psi_{n,k_y,k_z}(x, y, z) = e^{i(k_y y + k_z z)} \cdot \phi_n(x - x_0)$$

Different k_y values give same energy $\rightarrow$ **degeneracy**

Landau Level Degeneracy

The energy E_n does not depend on k_y (or x_0).

For a sample of area $A = L_x \times L_y$:

$$N = \frac{|q|BA}{h} = \frac{\Phi}{\Phi_0}$$

- $\Phi = BA$ = total magnetic flux
- $\Phi_0 = h/|q|$ = flux quantum
- Degeneracy $\propto$ magnetic field strength

Example: Magnitude of Degeneracy

Consider a sample of size $1\,\text{cm} \times 1\,\text{cm}$ in a field $B = 1.5\,\text{T}$ (typical MRI).

The degeneracy is:

$$N = \frac{|q|BA}{h} = \frac{(1.6 \times 10^{-19})(1.5)(10^{-4})}{6.63 \times 10^{-34}} \approx 3.6 \times 10^{10}$$

- About **36 billion states** in each Landau level!
- Huge degeneracy $\rightarrow$ Quantum Hall Effect

References

- **MIT L14.3 Particle in a constant magnetic field: Landau Levels**
  https://www.youtube.com/watch?v=Uux0VkKaoxY&t=18s
- **Lecture 5: Motion of a charged particle in a magnetic field**
  http://www.tcm.phy.cam.ac.uk/~bds10/aqp/lec5_compressed.pdf

Auxiliary: Why the Landau Gauge?

Two common gauge choices for $\vec{B} = B\hat{z}$:

- **Landau Gauge**: $\vec{A} = (0, Bx, 0)$
  - Breaks rotational symmetry
  - Leads to Cartesian separable solution
- **Symmetric Gauge**: $\vec{A} = \frac{B}{2}(-y, x, 0)$
  - Preserves rotational symmetry
  - Leads to solutions in polar coordinates

Both give the *same physics* (same energy levels, same observables).

Auxiliary: Magnetic Length 1

The magnetic length ℓ_B arises naturally from the harmonic oscillator analogy. characteristic length of a harmonic oscillator:

$$l = \sqrt{\frac{\hbar}{m\omega}}$$

Substitute the cyclotron frequency $\omega \to \omega_c = \frac{|q|B}{m}$:

$$\ell_B = \sqrt{\frac{\hbar}{m\omega_c}} = \sqrt{\frac{\hbar}{m\left(\frac{|q|B}{m}\right)}} = \sqrt{\frac{\hbar}{|q|B}}$$

Auxiliary: Magnetic Length 2

Question: How small is ℓ_B in practice?

Example: For a strong field $B = 1.5\,\mathrm{T}$ (MRI):

$$\ell_B \approx \sqrt{\frac{1.055 \times 10^{-34}}{1.6 \times 10^{-19} \times 1.5}} \approx 21\,\mathrm{nm}$$

Physical Meaning:

- Defines the size of the electron's wavefunction
- Area per quantum state: $A_{\text{state}} \approx 2\pi\ell_B^2$

Auxiliary: Quantum Hall Effect

When Landau levels are filled:

$$\sigma_{xy} = \nu \frac{e^2}{h}$$

- ν = number of filled Landau levels
- $R_K = h/e^2 \approx 25.8\,\mathrm{k}\Omega$ = von Klitzing constant
- Used as resistance standard (Nobel Prize 1985)